

Week 8: Residue Calculus and Applications

Order of a singularity

If $f(z)$ has an isolated singularity at z_0 then we can write a Laurent series:

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \frac{b_1}{z-z_0} + \frac{b_2}{(z-z_0)^2} + \dots$$

The sum of the singular terms is called the principal part of f .

We can classify the singular point based on the number of non-zero terms in the principal part.

Def: Suppose the Laurent series of a function f at z_0 has N non-zero terms. Then,

- if $N=0$ z_0 is a removable singularity
- if $N=\infty$ z_0 is an essential singularity
- if $1 \leq N < \infty$ z_0 is a pole of order m where $m \in \mathbb{N}$ has $b_m \neq 0$ and $b_{m+k} = 0$ for all $k \geq 1$.

A removable singularity can be removed by redefining f at z_0 .

Example: $f(z) = \frac{e^z - 1}{z}$ as a formula is not defined at $z=0$

however we saw $f(z) = 1 + \frac{1}{2!}z + \frac{1}{3!}z^2 + \dots$ and so we

can take $f(0) = 1$ making f entire.

The behaviour near poles of finite-order and essential singularities can be quite different. We will start with poles.

Residues at Poles

Poles with a finite order behave well in the sense that they locally look like an analytic function divided by a monomial with a root at the singularity.

This leads to a second criteria for a singularity involving analyticity and differentiation instead of integration and Laurent series.

Thm: An isolated singular point z_0 of a function f is a pole of order $-m$ if, and only if, $f(z)$ can be written as

$$f(z) = \frac{\phi(z)}{(z-z_0)^m} \quad (*)$$

where $\phi(z)$ is analytic and non-zero at $z=z_0$.

Moreover,

$$\operatorname{Res}_{z=z_0} f(z) = \begin{cases} \phi(z_0) & \text{if } m=1 \\ \frac{\phi^{(m-1)}(z_0)}{(m-1)!} & \text{if } m \geq 2. \end{cases}$$

Proof: " $\Leftarrow$ " Suppose that $f(z)$ has the form $(*)$. Then by analyticity of $\phi(z)$ it has a Taylor series:

$$\phi(z) = \phi(z_0) + \phi'(z_0)(z-z_0) + \dots + \frac{\phi^{(m-1)}(z_0)}{(m-1)!}(z-z_0)^{m-1} + \frac{\phi^{(m)}(z_0)}{m!}(z-z_0)^m + \dots$$

which converges for $|z-z_0| < R$. But then we can find the Laurent series for f at z_0 by division:

$$f(z) = \frac{\phi(z_0)}{(z-z_0)^m} + \frac{\phi'(z_0)}{(z-z_0)^{m-1}} + \dots + \frac{\phi^{(m-1)}(z_0)}{(m-1)!(z-z_0)} + \frac{\phi^{(m)}(z_0)}{m!} + \dots$$

which will converge for all $0 < |z-z_0| < R$. From this we conclude that z_0 is a singularity of order $-m$ and

$$\operatorname{Res}_{z=z_0} f(z) = \frac{\phi^{(m-1)}(z_0)}{(m-1)!}.$$

" $\Rightarrow$ " On the other hand, if f has a singularity of order $-m$ then

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \frac{b_1}{z-z_0} + \dots + \frac{b_m}{(z-z_0)^m}$$

for all $0 < |z-z_0| < R$. As a result we see that the function defined by

$$\phi(z) = \begin{cases} (z-z_0)^m f(z), & z \neq z_0 \\ b_m, & z = z_0 \end{cases}$$

is analytic as it has a convergent Taylor series given by

$$\phi(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^{n+m} + b_1 (z-z_0)^{m-1} + \dots + b_m.$$

□

Example: Compute $\int_C \frac{z^3+2z}{(z-i)^3} dz$ for C a circle around i oriented positively.

The integrand is of the form $\frac{\phi(z)}{(z-i)^3}$ where $z=i$ is the only singularity inside C .

$$\begin{aligned} \text{Hence, } \int_C \frac{z^3+2z}{(z-i)^3} dz &= \operatorname{Res}_{z=i} \frac{z^3+2z}{(z-i)^3} = \frac{1}{2!} \frac{d^2}{dz^2} (z^3+2z) \Big|_{z=i} \\ &= \frac{1}{2} (6z) \Big|_{z=i} = 3i \end{aligned}$$

Example. Compute $\int_C \frac{\log(z)}{z^2+1} dz$ where $\log(z) = \ln r + e^{i\theta}$ for $r > 0$, $\theta \in [0, 2\pi]$

and C is a circle of radius $\frac{1}{2}$ centered at $z=i$.

The integrand is of the form $\frac{\phi(z)}{z-i}$ where $\phi(z) = \frac{\log(z)}{z+i}$.

$$\text{Hence, } \operatorname{Res}_{z=i} f(z) = \phi(i) = \frac{\log(i)}{i+i} = \frac{i\pi/2}{2i} = \frac{\pi}{4}.$$

Example: what is the residue of $f(z) = \frac{\sinh z}{z^4}$ at $z=0$?

Note the answer is not given by $\frac{1}{3!} \phi^{(3)}(0)$.

We can compute the Laurent series:

$$\begin{aligned} f(z) &= \frac{1}{z^4} \left(z + \frac{1}{3!} z^3 + \frac{1}{5!} z^5 + \dots \right) \\ &= \frac{1}{z^3} + \frac{1}{3!} z + \frac{1}{5!} z^3 + \dots \end{aligned}$$

and so $z=0$ is a pole of order 3 not 4.

The reason for this is that $\sinh(z)$ is 0 at $z=0$. Instead, we need to write

$$f(z) = \frac{\sinh(z)/z}{z^3}$$

to correctly use this formula.

Another consequence of the above theorem is that the function has known behaviour near z_0 .

Thm: If z_0 is a pole of a function f then

$$\lim_{z \rightarrow z_0} f(z) = \infty.$$

Proof: By the previous theorem we can find a $\phi(z)$ and $m \geq 1$ so that

$$f(z) = \frac{\phi(z)}{(z-z_0)^m}$$

where $\phi(z_0) \neq 0$. Then

$$\lim_{z \rightarrow z_0} \frac{1}{f(z)} = \lim_{z \rightarrow z_0} \frac{(z-z_0)^m}{\phi(z)} = 0$$

which proves the claim. $\square$

Zeros of Analytic Functions

A lot of the time a function has a pole due to division by an analytic function which is zero.

We can understand poles by first understanding zeros.

If f is analytic at z_0 and $f(z_0) = 0$, then z_0 is a root or zero of f if $f^{(k)}(z_0) = 0$ for all $k = 0, 1, 2, \dots, m$.
then z_0 is a root of order m .

Note that:

$f(z)$ has a zero of order m at $z = z_0$ $\Leftrightarrow$ there exists an analytic function $g(z)$ s.t.
$$f(z) = (z - z_0)^m g(z)$$

Using this we can show that zeros of analytic functions are isolated (assuming f is not constant).

Thm: Suppose f is analytic at z_0 , $f(z_0) = 0$ and not identically zero in any neighbourhood of z_0 . Then $f(z) \neq 0$ for all z in the deleted neighbourhood $\{z : 0 < |z - z_0| < \varepsilon\}$.

Proof: By assumption we can find an m s.t. $f^{(m)}(z_0) \neq 0$.
Indeed, if not then the Taylor series for f at z_0 is identically zero, contradicting the assumption.

But then $f(z) = (z - z_0)^m g(z)$

where $g(z)$ is analytic at z_0 and $g(z_0) \neq 0$

Since g is continuous and non-zero, we know that $g(z) \neq 0$ for all z close enough to z_0 .

□

We conclude that zeroes of non-constant analytic functions are isolated. This also means that poles are also generally isolated.

Moreover, we can come up with formulas for the residue when a pole is formed by division of an analytic function.

Thm: Suppose p, q are analytic at z_0 , $p(z_0) \neq 0$ and z_0 is a simple pole of $q(z)$. Then

$$\operatorname{Res}_{z=z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q'(z_0)}.$$

Essential Singularities

Behaviour of functions nearby essential singularities is much less ordered.

Thm: (Casorati-Weierstrass)

Suppose that z_0 is an essential singularity of a function f , and let w_0 be any fixed complex number.

Then for any $\varepsilon > 0$, $\delta > 0$ you can find a point z in the deleted neighbourhood $\{z: |z - z_0| < \delta\}$ such that

$$|w_0 - f(z)| < \varepsilon.$$

That is, the function f fails to have any limit as $z \rightarrow z_0$ in a very dramatic way.

A stronger result, Piccard's Thm, actually shows that f takes all but 1 complex value infinitely often near z_0 .

We will not prove these results here, instead we will look at the example $f(z) = e^{1/z}$ on the homework.

Applications of Residues

Evaluation of Improper Integrals

Goal: Compute $\int_0^{\infty} f(x) dx := \lim_{R \rightarrow \infty} \int_0^R f(x) dx$.

$$\text{or } \int_{-\infty}^{\infty} f(x) dx := \lim_{R_1 \rightarrow -\infty} \int_{R_1}^0 f(x) dx + \lim_{R_2 \rightarrow \infty} \int_0^{R_2} f(x) dx.$$

When the limits exist (and are finite) we say the integrals converge.

However, sometimes we will refer to the Cauchy principal value of an integral

$$\text{P.V.} \int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_{-R}^R f(x) dx$$

which might exist even if the improper integral itself does not.

Example: P.V. $\int_{-\infty}^{\infty} x dx = 0$ but $\int_{-\infty}^{\infty} x dx$ does not exist.

Example: If $f(x)$ is an even function then if the Cauchy P.V. exists so does the full integral:

$$\begin{aligned} \int_0^R f(x) dx &= \frac{1}{2} \left(\int_0^R f(x) dx + \int_0^R f(x) dx \right) \\ &= \frac{1}{2} \left(\int_{-R}^0 f(y) dy + \int_0^R f(x) dx \right) \\ &\quad \begin{array}{l} \nearrow y = -x \Rightarrow dy = -dx \\ \text{and } x=R \Leftrightarrow y=-R \\ x=0 \Leftrightarrow y=0 \end{array} \\ &= \frac{1}{2} \int_{-R}^R f(y) dy \end{aligned}$$

$$\text{Hence, } \int_0^{\infty} f(x) dx = \frac{1}{2} \left\{ \text{P.V.} \int_{-\infty}^{\infty} f(y) dy \right\}.$$

Using the Cauchy-Residue Thm we can compute improper integrals of rational functions.

Suppose $f(z) = \frac{p(z)}{q(z)}$ for analytic p, q . Then the roots of $q(z)$ are poles of $f(z)$ (assuming $p(z) \neq 0$).

Assume that $q(z)$ has no real roots and at least one root above the axis. Moreover, assume that $q(z)$ has a finite number of zeroes. Then by letting C_R be the semi-circle of radius R centered at zero and above the real axis.

Then for large R

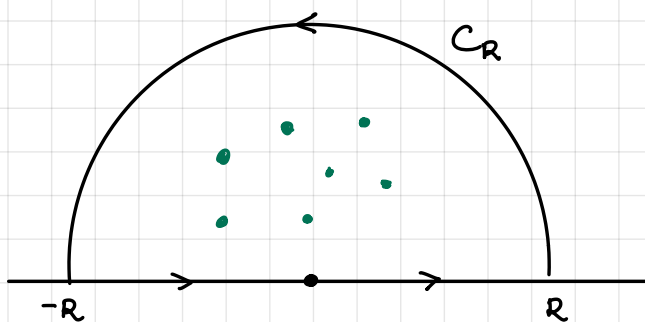
$$\int_{-R}^R f(x) dx + \int_{C_R} f(z) dz = 2\pi i \sum_{k=1}^N \operatorname{Res}_{z=z_k} f(z)$$

for the roots z_k of $q(z)$.

Assuming $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz$ exists, we conclude

$$\text{P.V.} \int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k} f(z) - \lim_{R \rightarrow \infty} \int_{C_R} f(z) dz$$

and if f is even this gives us the full improper integral.



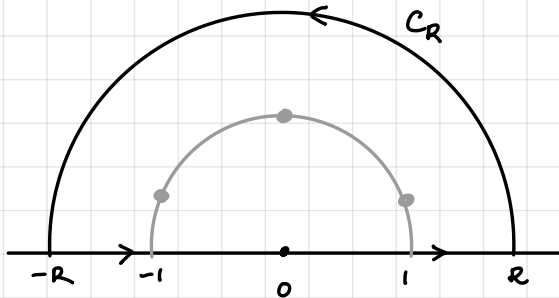
Example: $\int_{-\infty}^{\infty} \frac{x^2}{x^6+1} dx.$

Let $f(z) = \frac{z^2}{z^6+1}$ which is analytic away from the roots of the entire function z^6+1 .

These roots are the 6th-roots of -1 , which are

$$c_k = \exp\left[i\left(\frac{\pi}{6} + 2k\frac{\pi}{6}\right)\right] \quad \text{for } k = 0, 1, \dots, 5.$$

none of which lie on the real axis, and 3 of which are above the real axis. In particular, $c_0 = e^{i\pi/6}$, $c_1 = i$, $c_2 = e^{i5\pi/6}$.



By the Cauchy-Residue theorem:

$$\int_{-R}^R f(x) dx + \int_{C_R} f(z) dz = 2\pi i (B_0 + B_1 + B_2) \quad \text{--- (#)}$$

where B_k is the residue at pole c_k .

Each of the poles are simple (have order -1) and so we know

$$B_k = \operatorname{Res}_{z=c_k} f(z) = \frac{p(z_k)}{q'(z_k)} = \frac{c_k^2}{6c_k^5} = \frac{1}{6c_k^3}$$

$$\Rightarrow 2\pi i (B_0 + B_1 + B_2) = 2\pi i \left(\frac{1}{6i} + \frac{1}{-6i} + \frac{1}{6i} \right) = \frac{\pi}{3}$$

We can also show: $\left| \frac{z^2}{z^6+1} \right| \leq \frac{R^2}{R^6-1}$

$$\Rightarrow \left| \int_{C_R} f(z) dz \right| \leq \frac{R^2}{R^6-1} \cdot \pi R = \frac{\pi}{R^3 - 1/R^3} \longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Hence, $\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0.$

So by taking the limit as $R \rightarrow \infty$ in (#) that

$$\text{P.V.} \int_{-\infty}^{\infty} f(z) dz = \frac{\pi}{3} \quad (f(z) \text{ even})$$

which by evenness: $\int_{-\infty}^{\infty} f(z) dz = \frac{\pi}{3}$ and $\int_0^{\infty} f(z) dz = \frac{\pi}{6}.$

Improper Integrals from Fourier Analysis.

The Fourier transform of a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by:

$$\hat{f}(\xi) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{i\xi x} dx. \quad - (a)$$

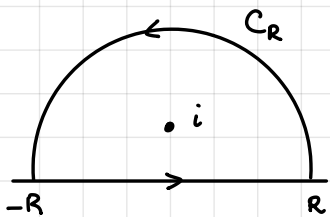
Taking real and imaginary parts, we get:

$$\int_{-\infty}^{\infty} f(x) \cos(\xi x) dx, \quad \int_{-\infty}^{\infty} f(x) \sin(\xi x) dx. \quad - (b)$$

We can use complex analysis to determine expressions for (a) and hence expressions for (b). However, note that our method only provides the value for the principal value, and this does not immediately tell us that the integral converges.

Example: $\int_{-\infty}^{\infty} \frac{\cos 3x}{(x^2+1)^2} dx$

Let $f(z) = \frac{1}{(z^2+1)^2}$ and note $f(z)e^{i3z}$ is analytic for $z \neq \pm i$.



Cauchy Residue Theorem says:

$$\int_{-R}^R \frac{e^{3xi}}{(x^2+1)^2} dx = 2\pi i \operatorname{Res}_{z=i} f(z)e^{i3z} - \lim_{R \rightarrow \infty} \int_{C_R} f(z)e^{i3z} dz \quad - (c)$$

The singularity at $z=i$ is order 2, but we have written

$$f(z)e^{i3z} = \frac{e^{i3z}}{(z^2+1)^2} = \frac{e^{i3z}}{(z+i)^2} \cdot \frac{1}{(z-i)^2} = \frac{\phi(z)}{(z-i)^2}$$

where $\phi(z) = \frac{e^{i3z}}{(z+i)^2}$ is analytic and not zero at $z=i$.

$$\begin{aligned} \Rightarrow \operatorname{Res}_{z=i} f(z)e^{i3z} &= \phi'(i) = \frac{d}{dz} \left(\frac{e^{i3z}}{(z+i)^2} \right) \Big|_{z=i} \\ &= \frac{(z+i)^2 \cdot 3ie^{i3z} - e^{i3z} \cdot 2(z+i)}{(z+i)^4} \Big|_{z=i} \end{aligned}$$

$$= \frac{(2i)^2 \cdot 3ie^{-3} - e^{-3} \cdot 2 \cdot 2i}{(2i)^4}$$

$$= \frac{(-12 - 4)ie^{-3}}{16} = \frac{1}{ie^3}.$$

Also we can again show that the C_R contribution goes to zero.

$$\left| \int_{C_R} f(z) e^{i3z} dz \right| \leq \pi R \cdot \frac{1}{|R^2 - 1|^2} \quad \text{where we use}$$

$$e^{i3z} = e^{i3x} e^{-3y}$$

$$\Rightarrow |e^{i3z}| = e^{-3y} \leq 1.$$

$$\longrightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Hence, we conclude that $\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{i3z} dz = 0.$

So taking the limit as $R \rightarrow \infty$ in (c) we get

$$\text{P.V.} \int_{-\infty}^{\infty} f(x) e^{i3x} dx = 2\pi i \cdot \frac{1}{ie^3} = \frac{2\pi}{e^3}.$$

Note that $\text{Re} \{ f(x) e^{i3x} \} = f(x) \cos(3x) = \frac{\cos 3x}{(x^2+1)^2}$

is an even function and so we also get

$$\int_{-\infty}^{\infty} \frac{\cos 3x}{(x^2+1)^2} dx = \frac{2\pi}{e^3}.$$

⚠ Warning: we do not get convergence of the imaginary part. Indeed,

$$\text{Im} \left\{ \text{P.V.} \int_{-\infty}^{\infty} \frac{\sin 3x}{(x^2+1)^2} dx \right\} = 0$$

But this is clear because by oddness

$$\int_{-R}^R \frac{\sin 3x}{(x^2+1)^2} dx = 0 \quad \text{for all } R$$

More work is needed to show that the imaginary part converges.

Jordan's Lemma

In both examples above we used that the contributions associated to C_R go to zero as $R \rightarrow \infty$.

We can guarantee this is the case for a large class of these Fourier type integrals.

Thm: (Jordan's Lemma)

Suppose $f(z)$ is :

- analytic in the upper half plane outside some half circle of radius R_0 , and
- for all points z on the circle C_R , there is a positive constant M_R s.t.

$$|f(z)| \leq M_R \quad \text{and} \quad \lim_{R \rightarrow \infty} M_R = 0$$

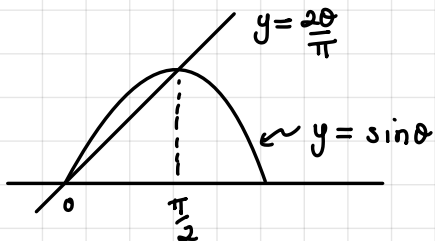
Then for every positive constant $a \in \mathbb{R}$

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{iaz} dz = 0.$$

Proof: The idea is to use Jordan's inequality

$$\int_0^\pi e^{-R \sin \theta} d\theta < \frac{\pi}{R}$$

which comes from the graph below and the fact e^x is an increasing function. Hence,



$$\begin{aligned} \sin \theta &\geq \frac{2\theta}{\pi}, \quad 0 \leq \theta \leq \pi/2 \\ \Rightarrow e^{-R \sin \theta} &\leq e^{-2\theta R/\pi}, \quad 0 \leq \theta \leq \pi/2 \\ \Rightarrow \int_0^{\pi/2} e^{-R \sin \theta} d\theta &\leq \int_0^{\pi/2} e^{-2\theta R/\pi} d\theta \\ &= \frac{\pi}{2R} (1 - e^{-R}) \\ &\leq \frac{\pi}{2R}. \end{aligned}$$

Now for the contour integral we have:

$$\int_{C_R} f(z) e^{iaz} dz = \int_0^\pi f(Re^{i\theta}) \exp(iaRe^{i\theta}) \cdot Rie^{i\theta} d\theta$$

$$\Rightarrow \left| \int_{C_R} f(z) e^{iaz} dz \right| \leq \int_0^\pi R M_R e^{-aR \sin \theta} d\theta \leq M_R \frac{\pi}{a}$$

Then as $M_R \rightarrow 0$ we find that $\lim_{R \rightarrow \infty} \int_{C_R} f(z) e^{iaz} dz = 0$. $\square$